

B.3 Gas laws

Topic warm-up · IB Physics (2023 syllabus)

with worked answers

Name: _____

Date: _____

Total: **25 marks**

$k_B = 1.38 \times 10^{-23} \text{ J K}^{-1}$. Show your working.

Question 1 — The angry fizzy bottle

(5 marks)

A sealed soft-drink bottle left in a hot car becomes noticeably harder to squeeze.

- (a) Explain, in terms of molecular collisions, what gas pressure is. [3]
- (b) Use the same molecular picture to explain why warming the sealed bottle raises the pressure. [2]

Question 2 — The bike pump and the beach day

(8 marks)

A cyclist inflates a tyre to 300 K in the morning shade. During the ride the tyre warms in the sun to 57 °C, at essentially constant volume.

- (a) Convert both temperatures to the scale the gas laws require, and state why that scale is the only valid choice. [3]
- (b) Calculate the factor by which the tyre pressure rises. [2]
- (c) A clubmate computes $57/27 = 2.1$, “more than double the pressure!”. Identify the error and state the correct percentage rise. [3]

Question 3 — Two changes at once

(5 marks)

A fixed mass of ideal gas has its absolute temperature doubled *and* its volume doubled.

- (a) Determine what happens to the pressure. [3]
- (b) A friend argues “both changes push the same way, so the pressure quadruples”. State which change the friend has misread, and what each change does to the pressure on its own. [2]

Question 4 — The heavyweight and the featherweight

(7 marks)

Two identical rigid boxes, at the same temperature of 300 K, hold equal numbers of molecules: one box of helium, one of xenon (about 33 times more massive per atom).

- (a) Compare the pressures in the two boxes, with justification from $pV = Nk_B T$. [2]
- (b) Calculate the average translational kinetic energy per molecule, and state for which box it applies. [3]
- (c) Hence state which gas has the faster-moving molecules, and by what reasoning. [2]

Answers

Question 1 — The angry fizzy bottle

- (a) Molecules are in constant random motion and collide with the container walls. Each collision reverses a molecule's momentum component normal to the wall, so the wall experiences a force. Pressure is the total rate of momentum transfer per unit area of wall, that is, force per unit area.
- (b) Warming raises the average molecular speed. Each collision then transfers more momentum, *and* collisions happen more often. Both effects increase the rate of momentum transfer per unit area, so the pressure rises. (The volume is fixed, so the wall area and the number of molecules are unchanged.)

Question 2 — The bike pump and the beach day

- (a) 300 K stays as it is, and $57^\circ\text{C} = 57 + 273 = 330\text{ K}$. The gas laws need a scale whose zero is the true zero of average molecular kinetic energy, so that $p \propto T$ holds. Only the kelvin scale has that property; the Celsius zero is an arbitrary point.
- (b) At constant volume $p \propto T$, so the pressure rises by a factor $330/300 = 1.1$.
- (c) The clubmate used Celsius values on both sides: 300 K is 27°C , and $57/27 = 2.1$. Because the Celsius zero is arbitrary, ratios of Celsius temperatures are meaningless. The correct rise is a factor of 1.1, that is 10%, not 110%.

Question 3 — Two changes at once

- (a) From $p = Nk_B T/V$: doubling T doubles p , while doubling V halves it. The two factors cancel exactly, so the pressure is *unchanged*.
- (b) The friend has misread the volume change. Increasing V alone *lowers* the pressure, because the same molecules strike a larger area less frequently; increasing T alone raises it. They oppose rather than reinforce.

Question 4 — The heavyweight and the featherweight

- (a) The pressures are *equal*. In $pV = Nk_B T$ the molecular mass does not appear, and N , V and T are identical for the two boxes.
- (b) $\bar{E}_k = \frac{3}{2}k_B T = 1.5 \times 1.38 \times 10^{-23} \times 300 = 6.2 \times 10^{-21}\text{ J}$. It applies to *both* boxes equally, since average translational kinetic energy depends only on temperature.
- (c) Helium. With the same average $\frac{1}{2}mv^2$ but a much smaller mass per atom, the mean speed must be larger, by a factor $\sqrt{33} \approx 5.7$.